# DeviceQoSEvaluator

## System Description

**Abstract**
This document describes the DeviceQoSEvaluator support system of the Eclipse Arrowhead 5.2.2 framework, as implemented in `ah5-DeviceQoSEvaluator-java-spring` of the primary Java source tree (`Eclipse-Arrowhead-java`) and deployed with the standalone Compose definition of that folder.

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# 1 Overview

This document describes the DeviceQoSEvaluator system. DeviceQoSEvaluator measures the devices that host the systems of the local cloud (round-trip time; CPU, memory and network load through device agents), keeps aggregated statistics and evaluates candidate providers for DynamicServiceOrchestration.




Figure 1: SysML-style block view of DeviceQoSEvaluator with its produced and consumed services.

The rest of this document is organized as follows. In Section 1.1, we reference major prior art capabilities of the system. In Section 1.2, we describe the intended usage of the system. In Section 1.3, we describe fundamental properties provided by the system. In Section 1.4, we describe delimitations of capabilities of the system. In Section 2, we describe the abstract service functions consumed or produced by the system. In Section 3, we describe the security capabilities of the system.

## 1.1 Significant Prior Art

New in Arrowhead 5 (AITIA 5.2.2 implementation).

## 1.2 How This System Is Meant to Be Used

DynamicServiceOrchestration calls `qualityEvaluation.filter` or `sort` when a consumer's request carries QoS requirements and `enable.qos=true`. Operators read statistics with `deviceQualityDataManagement`.

## 1.3 System functionalities and properties

### 1.3.1 Functional properties of the system

- Build the device-system map from the ServiceRegistry.
- Measure RTT and collect augmented metrics periodically; aggregate them.
- Filter and sort providers by weighted metrics over a time window.

### 1.3.2 Configuration of system properties

The system is configured with Spring properties (`application.properties`, overridden by environment variables in the container deployment). The properties that shape its behaviour are:

| Property | Default | Meaning |
|---|---|---|
| `server.port` | 8472 | HTTPS port. |
| `authentication.policy` | declared | Standalone deployment: `outsourced` with Authentication identity tokens and mutual TLS. |
| `management.policy` / `management.whitelist` | authorization | Access to the management services. |
| `enable.blacklist.filter` / `force.blacklist.filter` | false / true | Blacklist check of requesters. |
| `rtt.measurement.job.interval` / `rtt.measurement.timeout` | 10 s / 5000 ms | RTT measurement. |
| `augmented.measurement.job.interval` | 13 | Device-agent measurement interval (s). |
| `evaluation.time.window` | 60 | Default evaluation window (s). |
| `raw.measurement.data.max.age` / `inactive.device.max.age` | 15 / 30 | Retention (minutes). |
| `normalization.mode` | extended | Name normalization. |
| `mqtt.api.enabled` | false | Offer the MQTT interfaces. |
| `sysmonitor.enabled`, `sysmonitor.broker.*`, `sysmonitor.ssl.*` | true, amqp-broker:5671, server.ssl stores | Native `sysMonitor` publisher of the common library. |
| `arrowhead.ssl.crl-file` | (unset) | CRL checked for every client certificate (standalone deployment: `/crl/ca.crl`). |

Secrets (database, keystore and identity passwords, secret keys) have no usable defaults; the standalone deployment reads them from per-deployment Compose file secrets under `/run/secrets`.

The standalone deployment of this folder uses `config/application.properties` (copied into the image) with `server.port`=8472, `authentication.policy=outsourced`, `service.interface.policy=CERT_AUTH`, `management.policy=sysop-only`, `service.authorization.enabled=true`, `mqtt.api.enabled=false`, `arrowhead.ssl.crl-file=/crl/ca.crl`; secrets are read from Compose file secrets under `/run/secrets` (`spring.config.import=configtree:/run/secrets/`). `enable.blacklist.filter` is not enabled there, so incoming requests are not checked against the Blacklist; blacklisted providers are still removed by DynamicServiceOrchestration.

### 1.3.3 Data stored by the system

Devices, systems and statistic tables per metric group, and log entries (database `ah_device_qos_evaluator`).

### 1.3.4 Non functional properties

Measurement jobs run on a Quartz thread pool (60 threads).

### 1.3.5 Stateful or stateless

Stateful: measurements and statistics are persistent for their retention.

## 1.4 Important Delimitations

Only measured devices can be evaluated; augmented metrics require device agents.

# 2 Services

This section describes the produced and consumed services. Each produced service is specified by an SD document and one IDD document per interface.

## 2.1 Produced services

| Service | Version | Interfaces | SD | IDD |
|---|---|---|---|---|
| `qualityEvaluation` | 1.0.0 | HTTP(S), MQTT(S) | [1] | [2], [3] |
| `deviceQualityData Management` | 1.0.0 | HTTP(S), MQTT(S) | [4] | [5], [6] |
| `generalManagement` | 1.0.0 | HTTP(S), MQTT(S) | [7] | [8], [9] |
| `sysMonitor` | 1.0.0 | AMQP(S) | [10] | [11] |

The HTTP interfaces are always offered, the MQTT interfaces with `mqtt.api.enabled=true` and the AMQP `sysMonitor` interface with `sysmonitor.enabled=true`. The unregistered endpoints of the implementation are: `GET /deviceqosevaluator/monitor/echo`.

## 2.2 Consumed services

| Provider | Service | Operations | Use | Condition |
|---|---|---|---|---|
| ServiceRegistry | systemDiscovery, serviceDiscovery | register, revoke, lookup | Registration of the system and its services; lookup of the bootstrap services. | always |
| ServiceRegistry | serviceRegistry Management | system-query, device-query | Device-system map. | management whitelist |
| Authentication | identity | identity-login, identity-verify | Own identity token; verification of requester tokens. | `authentication. policy= outsourced` |
| Consumer Authorization | authorization | verify | Management filter decisions. | `management. policy= authorization` |
| Blacklist | blacklistDiscovery | check | Blacklist filter. | `enable. blacklist. filter=true` |
| Device agents | device metrics | query | Augmented metrics. | agents deployed |
| AMQP broker (SysMonitor) | sysMonitor (publisher) | echo, power, status | Own observations. | `sysmonitor. enabled=true` |

Except for the bootstrap services, which are looked up in the ServiceRegistry, the system reaches other systems' services through DynamicServiceOrchestration (`http.collector.mode=ORCHESTRATION`, results cached for 300 s); every such call needs a ConsumerAuthorization grant.

# 3 Security

The system can run in an insecure mode (`server.ssl.enabled=false`, plain HTTP, identity declared by the requester) for development, and in the Arrowhead secure mode with HTTPS and X.509 certificates following the Arrowhead certificate profile (`<System>.<Cloud>.<Organization>.arrowhead.eu`). Certificates not issued under the cloud CA are refused by the TLS handshake. The standalone deployment of this folder (`config/application.properties` and `compose.yaml`) runs the secure mode.

## 3.1 Security model

- **Protocols:** HTTPS (TLS 1.3 and 1.2) with mandatory client certificates; MQTT over TLS; AMQP over TLS for `sysMonitor`.
- **Data protection:** transport encryption by TLS; the database connection uses TLS with server verification in the standalone deployment. Secrets are kept out of the container environment (Compose file secrets).
- **System authentication:** by the authentication policy: `certificate`, `outsourced` (identity token verified with Authentication) or `declared` (insecure); the standalone deployment uses `outsourced`. An identity token is accepted only together with the client certificate of the system it was issued to (token-to-certificate binding), and every client certificate is checked against the cloud CA's CRL (fail closed).
- **Authorization of produced services:** management services are protected by the management filter (`management.policy`: `SYSOP_ONLY`, `WHITELIST` or `AUTHORIZATION`); with `service.authorization.enabled=true` ConsumerAuthorization decides each call; an optional blacklist filter refuses blacklisted requesters. Sysop identities bypass the management filter.
- Needs ServiceRegistry management access (whitelist and direct access) to build its map.

For the Arrowhead certificate profile see github.com/eclipse-arrowhead/documentation.

# 4 References

[1] J. Delsing, "qualityEvaluation – Service Description (DeviceQoSEvaluator)," Eclipse Arrowhead, SD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-qualityEvaluation-SD.pdf.

[2] ——, "qualityEvaluation HTTP/TLS/JSON – Interface Design Description (DeviceQoSEvaluator)," Eclipse Arrowhead, IDD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-qualityEvaluation-HTTP-IDD.pdf.

[3] ——, "qualityEvaluation MQTT/TLS/JSON – Interface Design Description (DeviceQoSEvaluator)," Eclipse Arrowhead, IDD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-qualityEvaluation-MQTT-IDD.pdf.

[4] ——, "deviceQualityDataManagement – Service Description (DeviceQoSEvaluator)," Eclipse Arrowhead, SD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-deviceQualityDataManagement-SD.pdf.

[5] ——, "deviceQualityDataManagement HTTP/TLS/JSON – Interface Design Description (DeviceQoSEvaluator)," Eclipse Arrowhead, IDD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-deviceQualityDataManagement-HTTP-IDD.pdf.

[6] ——, "deviceQualityDataManagement MQTT/TLS/JSON – Interface Design Description (DeviceQoSEvaluator)," Eclipse Arrowhead, IDD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-deviceQualityDataManagement-MQTT-IDD.pdf.

[7] ——, "generalManagement – Service Description (DeviceQoSEvaluator)," Eclipse Arrowhead, SD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-generalManagement-SD.pdf.

[8] ——, "generalManagement HTTP/TLS/JSON – Interface Design Description (DeviceQoSEvaluator)," Eclipse Arrowhead, IDD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-generalManagement-HTTP-IDD.pdf.

[9] ——, “generalManagement MQTT/TLS/JSON – Interface Design Description (DeviceQoSEvaluator),” Eclipse Arrowhead, IDD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-generalManagement-MQTT-IDD.pdf.

[10] ——, “sysMonitor – Service Description (DeviceQoSEvaluator),” Eclipse Arrowhead, SD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-sysMonitor-SD.pdf.

[11] ——, “sysMonitor AMQP/TLS/JSON – Interface Design Description (DeviceQoSEvaluator),” Eclipse Arrowhead, IDD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-sysMonitor-AMQP-IDD.pdf.

# 5 Revision History

## 5.1 Amendments

| No. | Date | Version | Subject of Amendments | Author |
|---|---|---|---|---|
| 1 | 2026-10-22 | 5.2.2 | Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace. | Jerker Delsing |
| 2 | 2026-10-30 | 5.2.2 | Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated. | Claude Opus 5.5 (draft) |

## 5.2 Quality Assurance

| No. | Date | Version | Approved by |
|---|---|---|---|
| 1 | | 5.2.2 | |